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90

TRANSPORT PHENOMENA II, CHE 312

QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

A liquid with a plug flow velocity profile $v_z(r) = v_0$ is being heated within a cylindrical tube of radius R with a constant heat flux $q_r(z, r = R) = q_R < 0$.

$$T_b = \frac{\int_S ds\, T(r,z) v_z}{\int_S ds\, v_z}$$

Model assumption:

- Steady-State
- Constant physical properties $\rho,\ c_p,\ k$.
- $c_V \approx c_P$ is constant and $\frac{du}{dT} = c_P$
- Heat flux in the axial direction is negligible.
- Negligible viscous heating.
- Fluid enters the heating region of the tube with a uniform temperature $T(z = 0, r) = T_1$ ← Inlet temp is uniform

$$\int_S ds\, \rho u \langle v | n \rangle + \int_S ds \langle q | n \rangle = 0$$

Derive an expression for the bulk (cup-mixing average) temperature of the fluid as a function of z.

$$\rho C_p \int_S ds\, v_z T(z=0,r) - \rho C_p \int_S \underbrace{ds\, v_z T(r,z)}_{= T_b \int_S ds\, v_z} + \underbrace{\int_S ds\, q_r}_{} = 0$$

↳ Constant heat flux so $q_r(r=R) = q_R$

$$\left\{ \begin{array}{l} u = C_p(\Delta T) \\ q_r = K \frac{dT}{dr} \leftarrow \text{Constant} \end{array} \right.$$

↳ plug this in

Inlet = $T_1 - T_0$ room temperature

outlet = $T_b - T_0$

$v_z(r) = v_0$

$$[\pi r^2 \rho C_p (T_1 - \cancel{T_0}) v_0] - [\pi r^2 \rho C_p (T_b - \cancel{T_0}) v_0] + 2\pi r q_R$$

$$v_0 \pi r^2 \rho C_p (T_1 - T_b) + 2\pi r \rho q_R = 0$$

↳ Solve for bulk temp

$$\frac{v_0 \pi r^2 C_p (T_1 - T_b)}{v_0 \pi r^2 C_p} = \frac{-2\pi r \rho q_R}{v_0 \pi r^2 C_p}$$

$$-T_b = -\frac{2\cancel{\pi}\cancel{r} \rho q_R}{v_0 \cancel{\pi} r^{\cancel{2}} C_p} - T_1$$

$$\boxed{T_b = T_1 + \frac{2\rho q_R}{v_0 r C_p}}$$

r - ok
z - very close
maths ok
interpretation close, missing z